



## Grade: Grade 9 · Science

### Climate Change & Ecosystem Impacts – Home Practice

Focus: Ecosystems · Food Web Changes · Climate Impact · Critical Thinking

#### ***Reading Passage:***

Ecosystems are delicate systems where living and non-living components depend on one another to survive. Climate change disrupts these systems by altering temperatures, rainfall patterns, and seasonal cycles. For example, warming oceans can cause coral bleaching, affecting fish populations that depend on reefs for shelter. In the Arctic, melting ice reduces hunting grounds for polar bears and shifts food availability. These changes create ripple effects, or chain reactions, where one change leads to many others. Understanding these patterns helps scientists predict future ecological challenges.

1. Identify two ways climate change disrupts ecosystems according to the passage.

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2. Explain what a 'chain reaction' means in the context of climate change.

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3. Describe how coral bleaching affects marine food webs.

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4. How does melting Arctic ice impact predators such as polar bears?

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5. List one biotic and one abiotic factor mentioned or implied in the passage.

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6. Create a short cause → effect → chain reaction sequence based on any example in the passage.

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7. Write a 2–3 sentence summary explaining how climate change affects entire ecosystems, not just one species.

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**Tutoring Tip:**

Think in sequences: one change in climate can cause many linked consequences in an ecosystem.